

CoRoT-2b

R-2149

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_120919 · created 2026-08-12T00:58:02+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456189.691 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.223 +/- 0.015
Transit depth (Rp/Rs)^2	4.98 +/- 0.65 [%]
Orbital Inclination (inc)	89.72 +/- 1.21
Airmass coefficient 1 (a1)	1.031 +/- 0.036
Airmass coefficient 2 (a2)	-0.015 +/- 0.019
Scatter in the residuals of the lightcurve fit is	3.56 %
Best Comparison Star	#2 - [153, 214]
Optimal Aperture	2.3
Optimal Annulus	11.58
Transit Duration (day)	0.1011 +/- 0.0016

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.223 ± 0.015 vs 0.1667 (deviation 2.61σ)
Duration fit vs published	0.1011 d vs 0.095 d (deviation 2.65σ)
Mid-time O–C	-3.0 min
Depth SNR	10.3
Residual scatter	3.56 %

Light curve

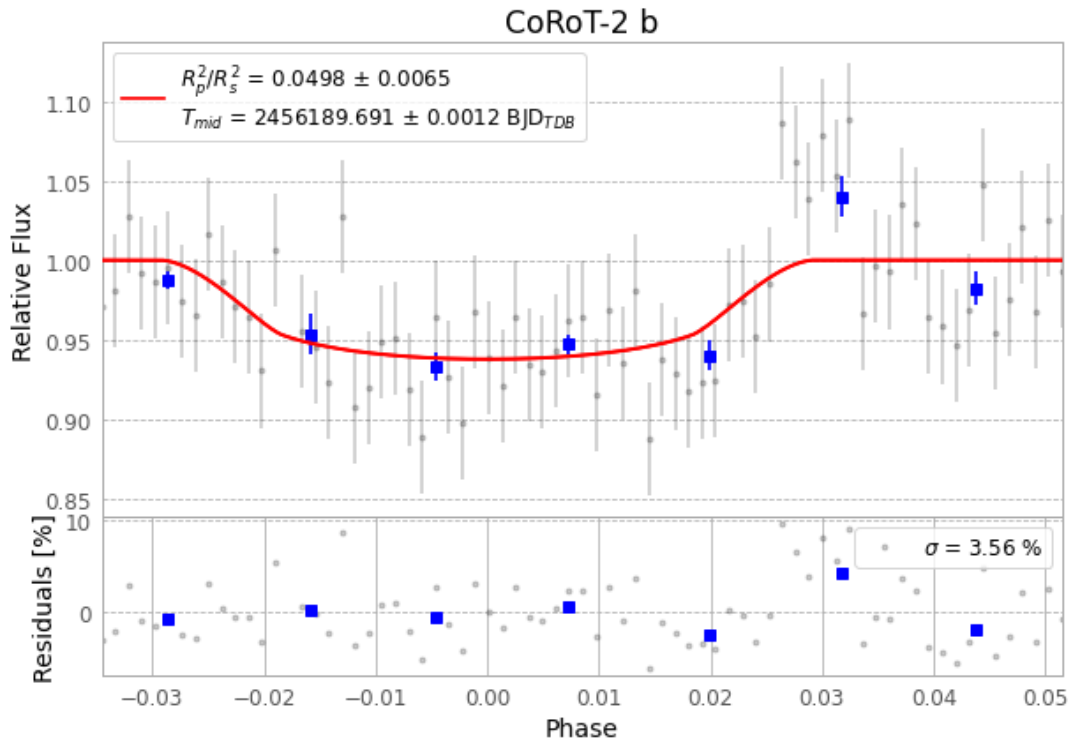


Figure 1 — FinalLightCurve_CoRoT-2 b_2012-09-18.png (SHA-256 354e288c20728608...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 274], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1781296e9c28c962...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

73 FITS frames (48 MB), CoRoT-2120919030313.FITS → CoRoT-2120919063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-120919.log (749ba074b1a2...), FinalLightCurve_CoRoT-2 b_2012-09-18.png (354e288c2072...), FinalLightCurve_CoRoT-2 b_2012-09-18.csv (34587178b958...)

Compute resources

Wall time 146.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:58Z.